SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Traversal Challenge

Course Code:	CSA0309	**Course Title:**	Data Structures
CO Assessed:	CO3 – Precision (BL3, BL5)	**Assessment:**	Assessment Tool 2 – Traversal Challenge
Weightage:	30% of CIA	**Total Marks:**	100
CO3 Statement:	*Solve problems for optimized data storage and retrieval using Binary Search Trees, AV Trees, BTrees, Red-Black Trees, and Splay Trees.*		
Student Name:	GAUTHAM G	**Reg. No.:**	192524371
Section / Batch:	SLOT-B	**Date:**	31/07/20

Code of Conduct

I am GAUTHAM G (192524371) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will re disciplinary action. Signature of the Student: GAUTHAM G

Instructions

- This assessment tool contains 5 Traversal Challenge questions, Total: 100 Marks.
- Students can use any online/offline Traversal Challenge. Demonstrate tree.
- When a student answer using simulator, in evaluation the answer should have captured screenshots after every major operation
- Submit the report in PDF format with properly labelled screenshots as proof of completion.

Section / Topic	Focus	Q.Nos	Marks
Tree Traversals	Traversal types, In order, Post order and Pre order	5	100
	GRAND TOTAL:		100 Marks

Traversal Challenge

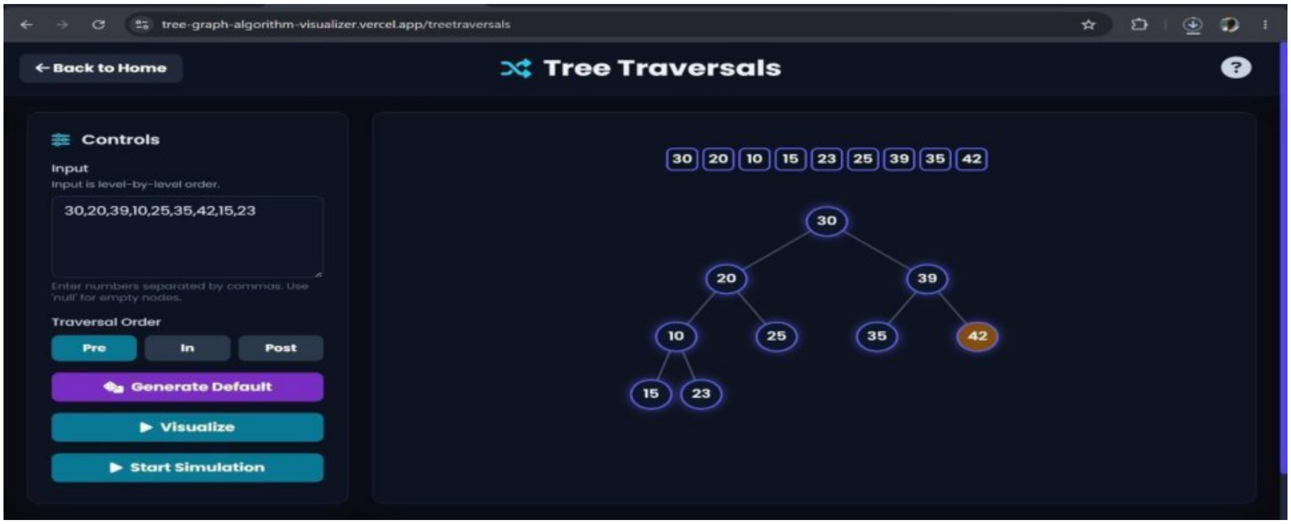
Rubrics for Calculation

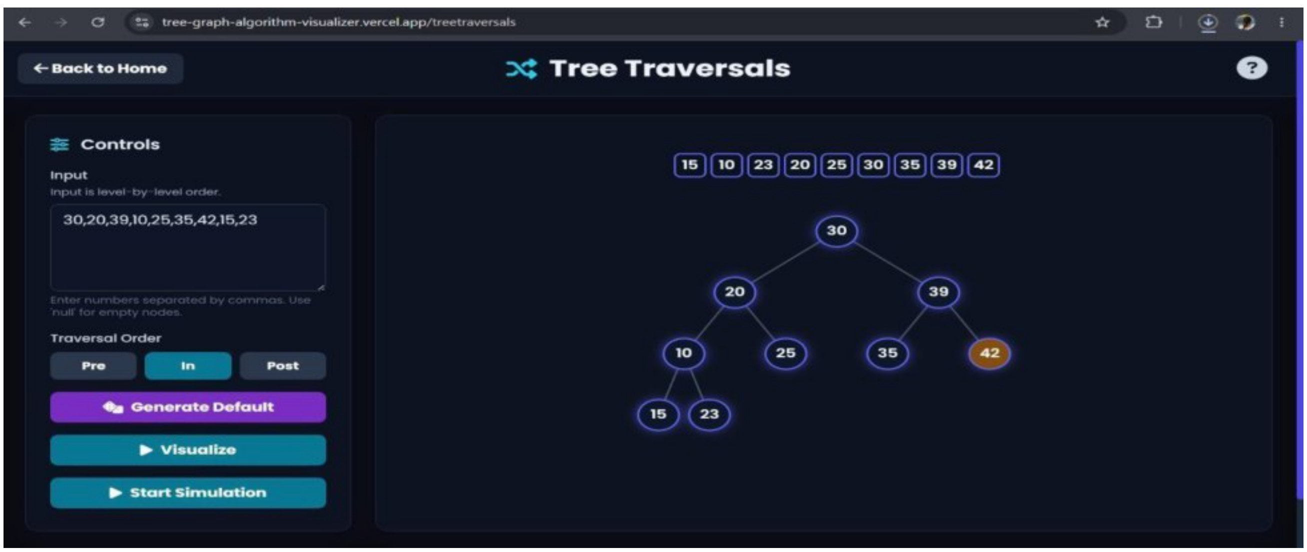
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning

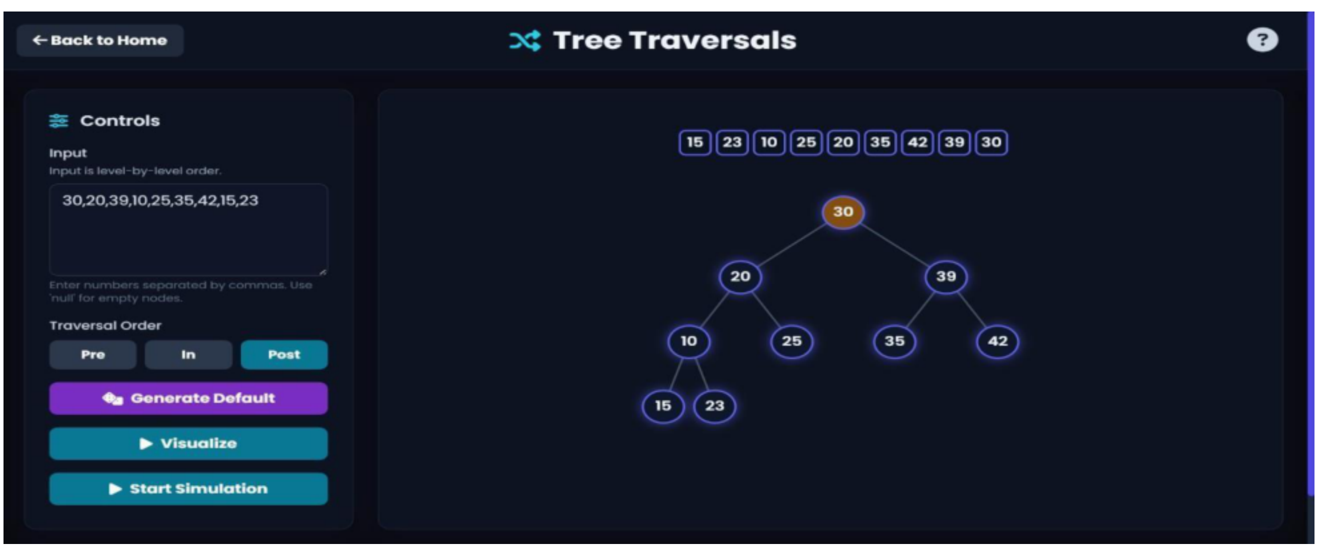
Conceptual Understanding	20	Demonstrates thorough understanding of all core Data Structures concepts; answers accurately reflect deep knowledge	Shows good understanding with minor conceptual gaps	Basic understanding; several misconceptions present	Limited understanding; major concepts misunderstood
Traversal Accuracy	20	Correctly completes all tree traversals (Preorder, Inorder, Postorder, Level Order) without errors.	Completes most traversals correctly with one minor error.	Correctly performs only some traversals with multiple errors.	Unable to perform most traversals correctly.
Selection of Appropriate Traversal	30	Correctly selects the appropriate traversal for every given problem scenario.	Selects the correct traversal for most scenarios.	Selects appropriate traversal for only a few scenarios.	Frequently selects incorrect traversal methods.
Completion & Time Efficiency	20	Completes all challenges within the allotted time while maintaining accuracy.	Completes most challenges within the time limit.	Completes only part of the challenge or exceeds the time limit slightly.	Unable to complete the challenge within the allotted time.
Evidence Submission	10	Uploads complete screenshots showing successful completion, score, and student details.	Uploads screenshots with minor missing information.	Uploads incomplete or unclear screenshots.	No valid evidence submitted.

Faculty In-charge	Course Coordinator	Program Director
Signature:	Signature:	Signature:
Date:	Date:	Date:

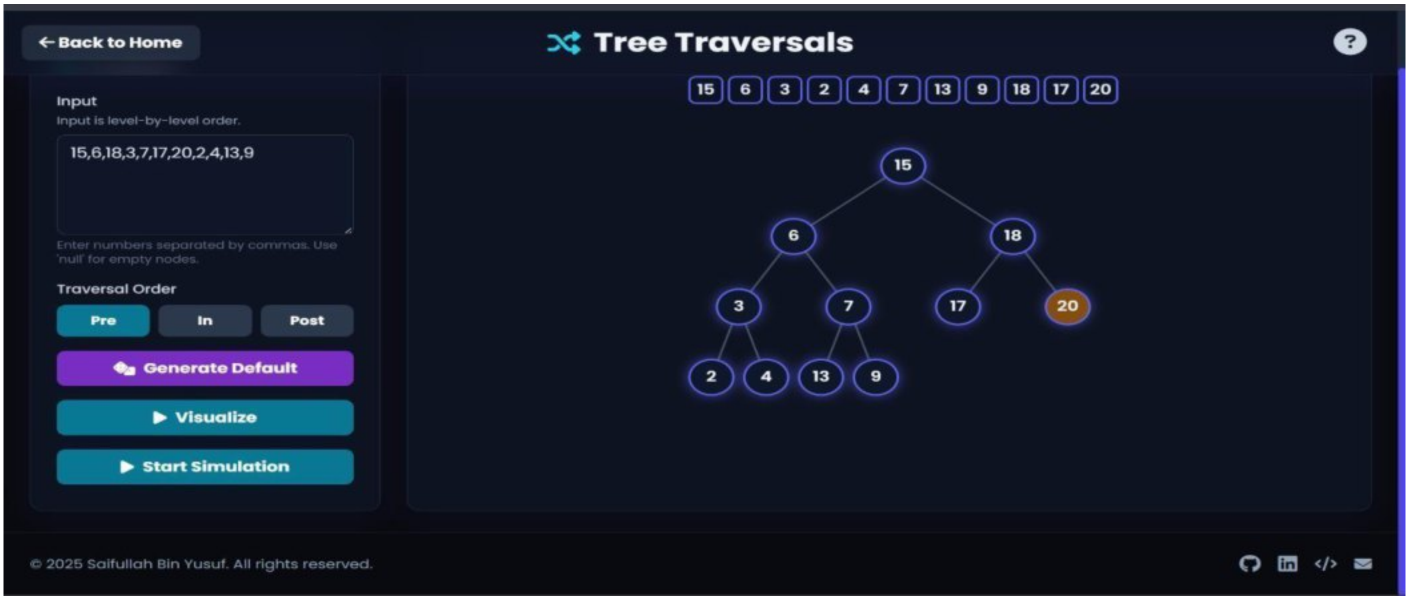
Question-1


tree-graph-algorithm-visualizer.vercel.app/treetraversals
Back to Home
Tree Traversals
Controls
Input
Input is level-by-level order.
30,20,39,10,25,35,42,15,23
Enter numbers separated by commas. Use 'null' for empty nodes.
Traversal Order
Pre
In
Post
Generate Default
Visualize
Start Simulation
30 20 10 15 23 25 39 35 42
30
20
39
10
25
35
42
15
23

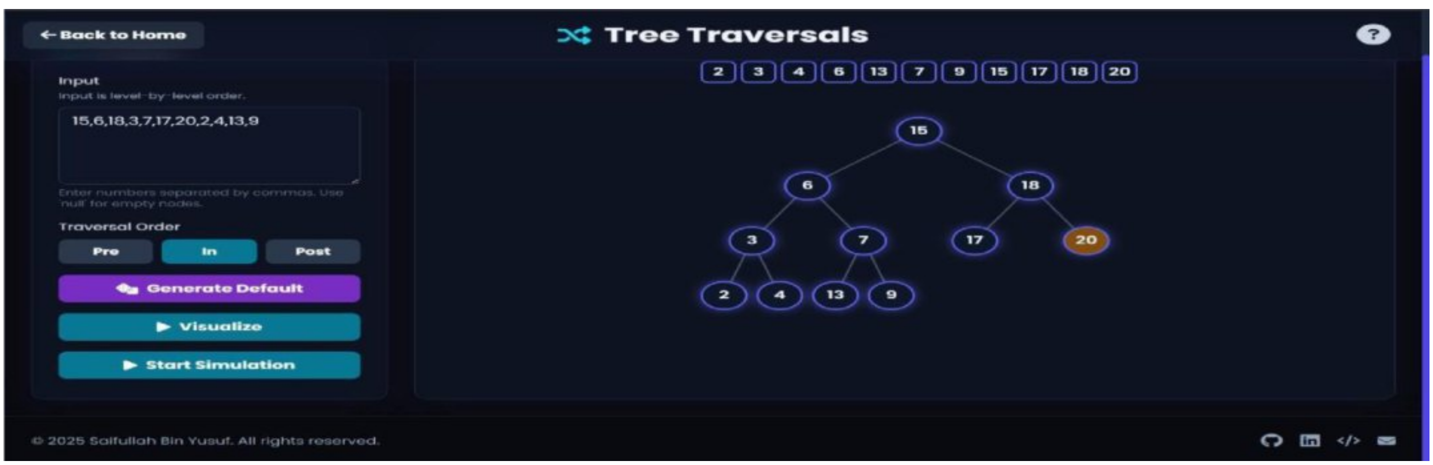

tree-graph-algorithm-visualizer.vercel.app/treetraversals
Back to Home
Tree Traversals
Controls
Input
Input is level-by-level order.
30,20,39,10,25,35,42,15,23
Enter numbers separated by commas. Use 'null' for empty nodes.
Traversal Order
Pre
In
Post
Generate Default
Visualize
Start Simulation
15 10 23 20 25 30 35 39 42
30
20
39
10
25
35
42
15
23


Back to Home
Tree Traversals
Controls
Input
Input is level-by-level order.
30,20,39,10,25,35,42,15,23
Enter numbers separated by commas. Use 'null' for empty nodes.
Traversal Order
Pre
In
Post
Generate Default
Visualize
Start Simulation
15 23 10 25 20 35 42 39 30
30
20
39
10
25
35
42
15
23

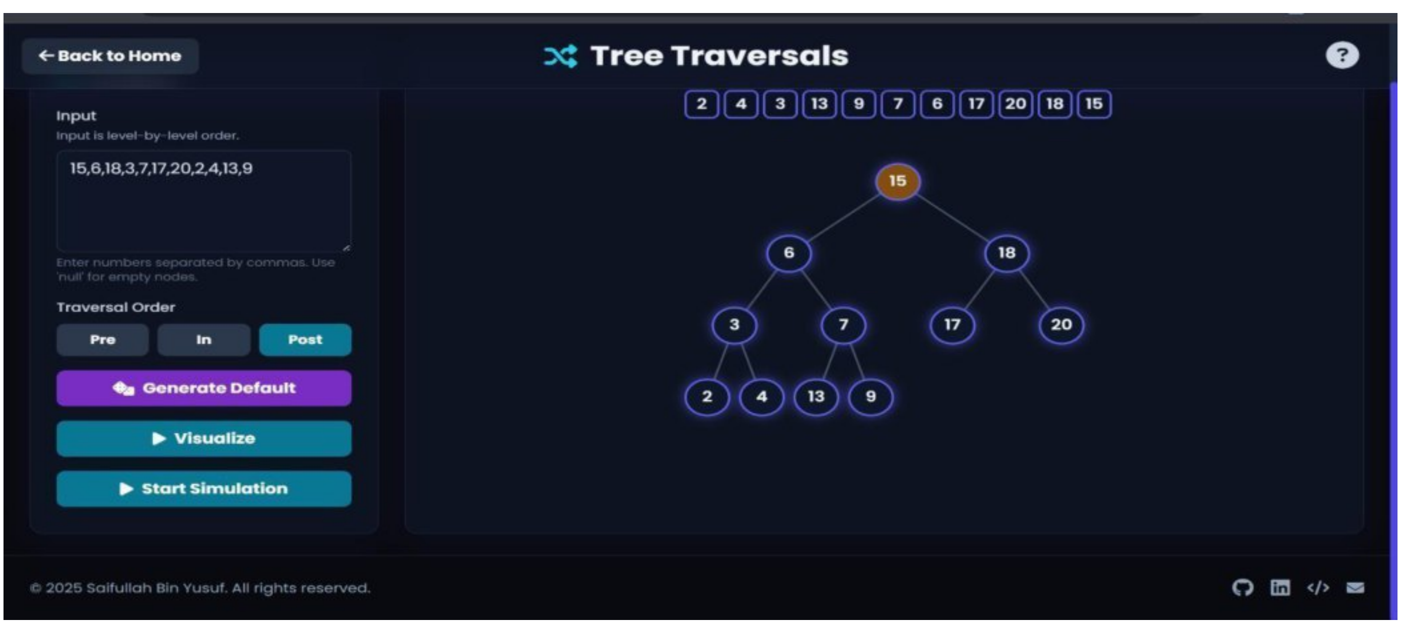
Question-2



Back to Home
Tree Traversals
15 6 3 2 4 7 13 9 18 17 20
Input
Input is level-by-level order.
15,6,18,3,7,17,20,2,4,13,9
Enter numbers separated by commas. Use 'null' for empty nodes.
Traversal Order
Pre
In
Post
Generate Default
Visualize
Start Simulation
15
6
18
3
7
17
20
2
4
13
9
© 2025 Saifullah Bin Yusuf. All rights reserved.

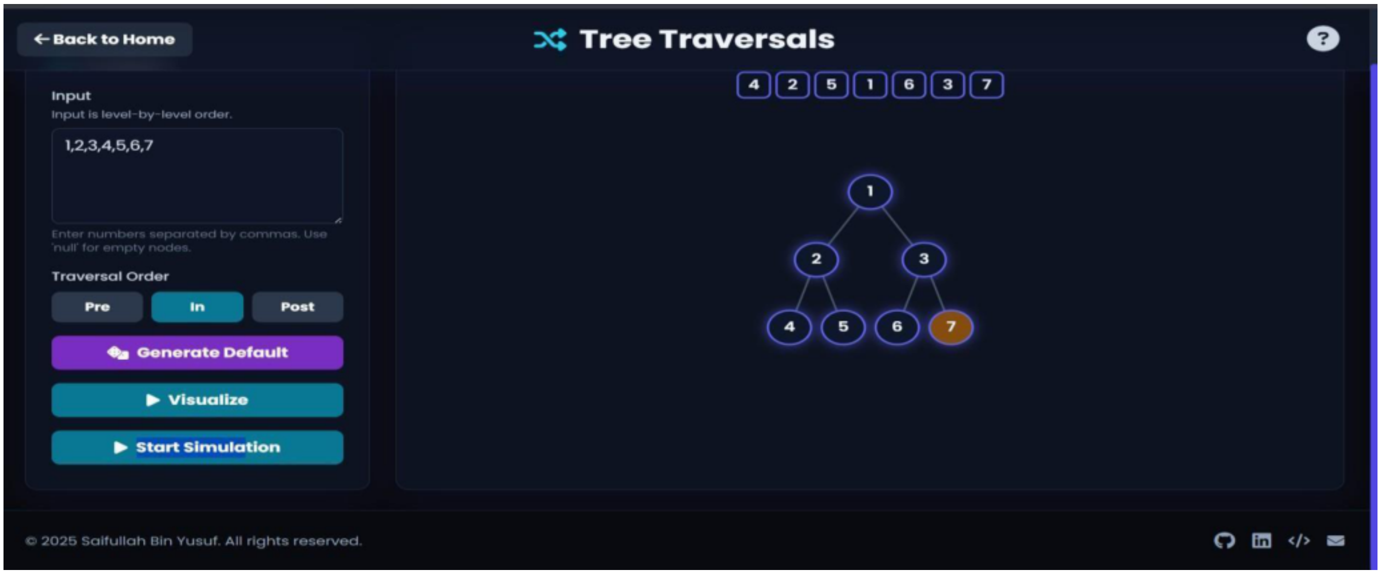


Back to Home
Tree Traversals
2 3 4 6 13 7 9 15 17 18 20
Input
Input is level-by-level order.
15,6,18,3,7,17,20,2,4,13,9
Enter numbers separated by commas. Use 'null' for empty nodes.
Traversal Order
Pre
In
Post
Generate Default
Visualize
Start Simulation
15
6
18
3
7
17
20
2
4
13
9
© 2025 Saifullah Bin Yusuf. All rights reserved.

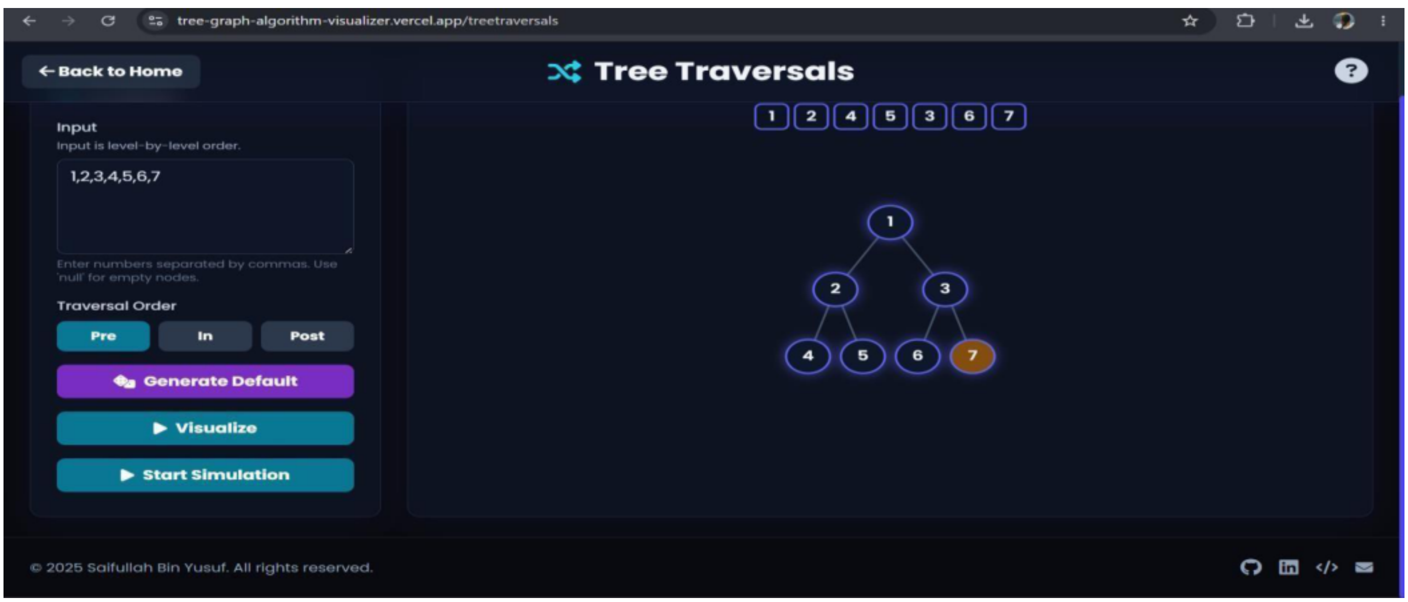


Back to Home
Tree Traversals
2 4 3 13 9 7 6 17 20 18 15
Input
Input is level-by-level order.
15,6,18,3,7,17,20,2,4,13,9
Enter numbers separated by commas. Use 'null' for empty nodes.
Traversal Order
Pre
In
Post
Generate Default
Visualize
Start Simulation
15
6
18
3
7
17
20
2
4
13
9
© 2025 Saifullah Bin Yusuf. All rights reserved.

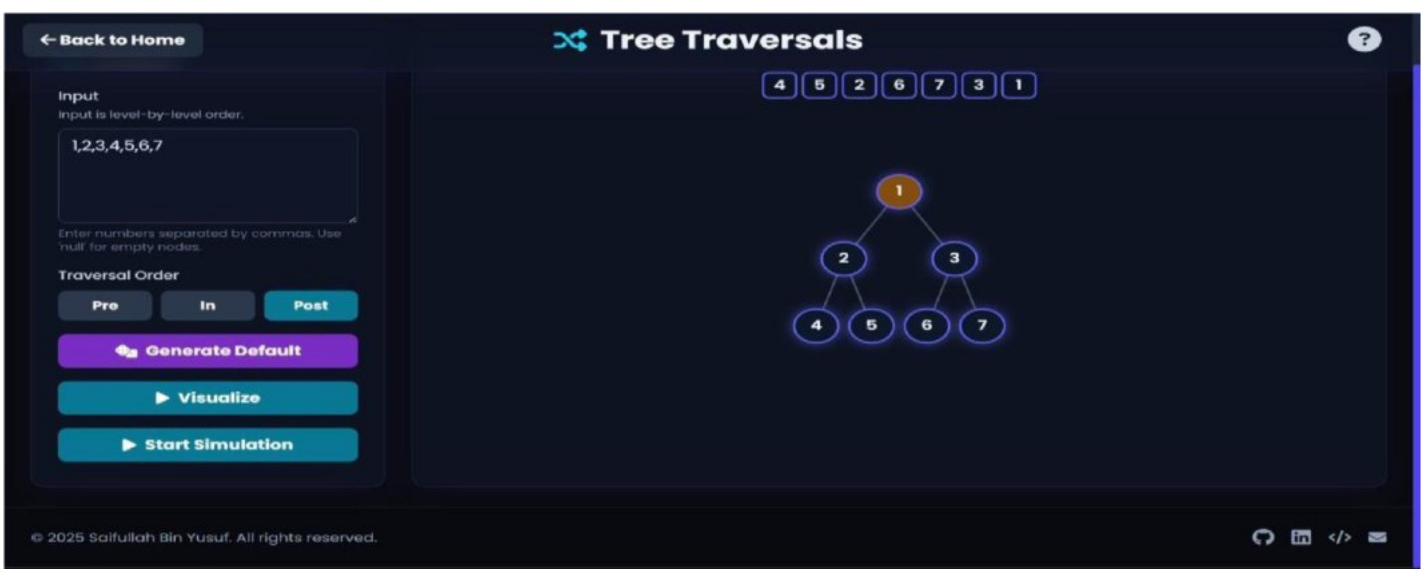
Question-3



← Back to Home
Tree Traversals
4 2 5 1 6 3 7
Input
Input is level-by-level order.
1,2,3,4,5,6,7
Enter numbers separated by commas. Use 'null' for empty nodes.
Traversal Order
Pre
In
Post
Generate Default
Visualize
Start Simulation
© 2025 Saifullah Bin Yusuf. All rights reserved.

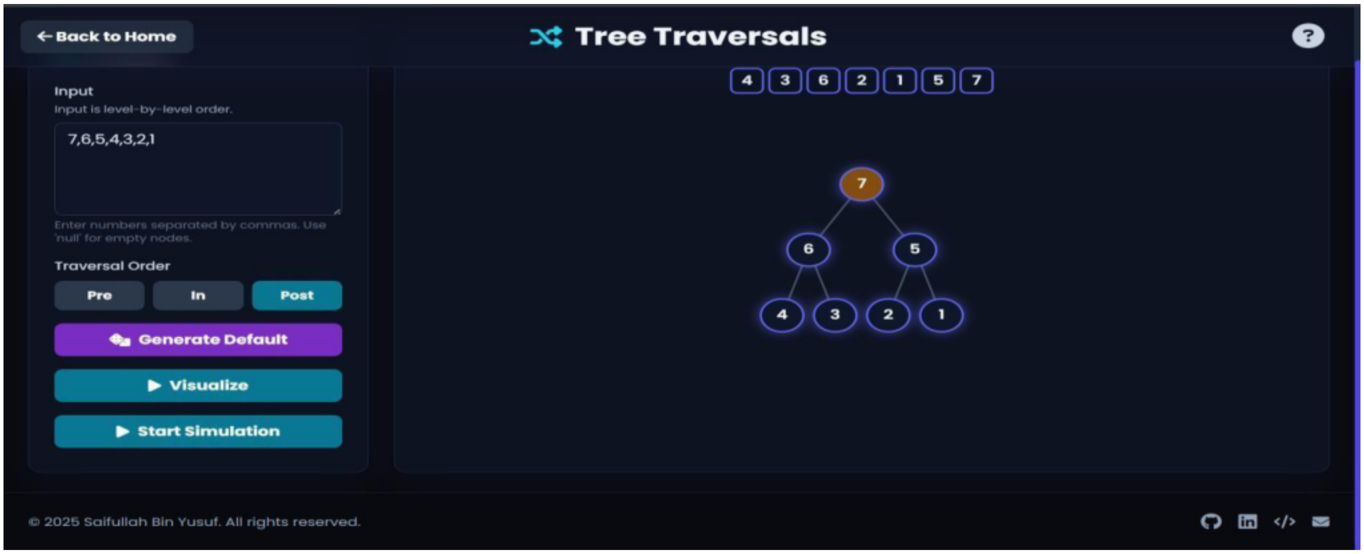


tree-graph-algorithm-visualizer.vercel.app/treetraversals
← Back to Home
Tree Traversals
1 2 4 5 3 6 7
Input
Input is level-by-level order.
1,2,3,4,5,6,7
Enter numbers separated by commas. Use 'null' for empty nodes.
Traversal Order
Pre
In
Post
Generate Default
Visualize
Start Simulation
© 2025 Saifullah Bin Yusuf. All rights reserved.

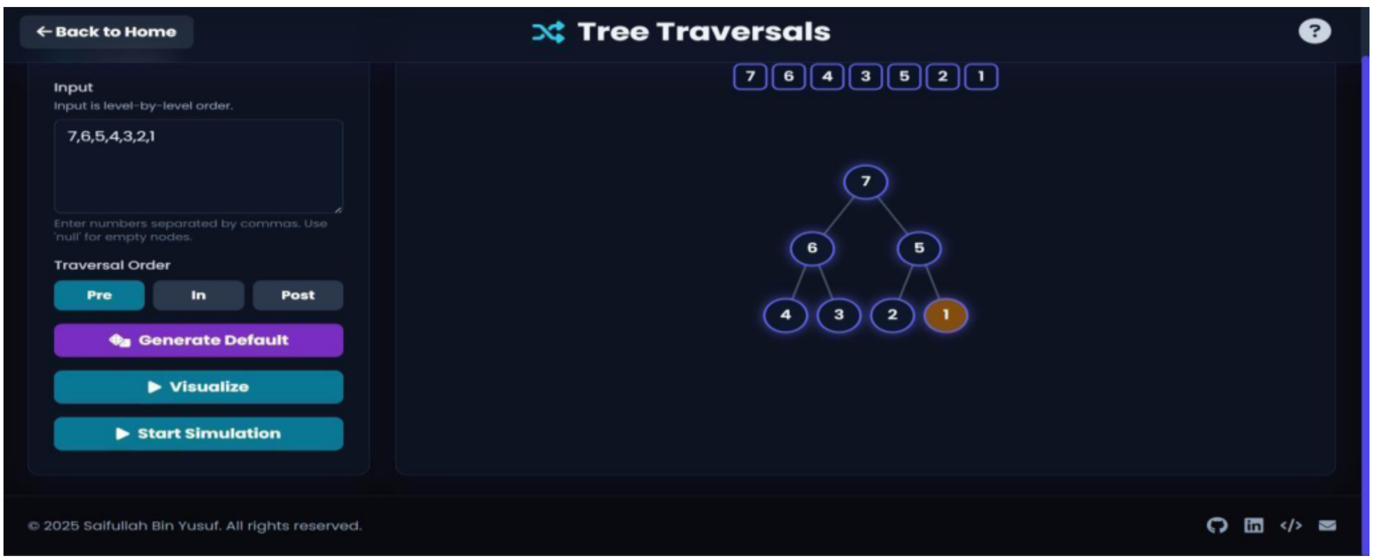


← Back to Home
Tree Traversals
4 5 2 6 7 3 1
Input
Input is level-by-level order.
1,2,3,4,5,6,7
Enter numbers separated by commas. Use 'null' for empty nodes.
Traversal Order
Pre
In
Post
Generate Default
Visualize
Start Simulation
© 2025 Saifullah Bin Yusuf. All rights reserved.

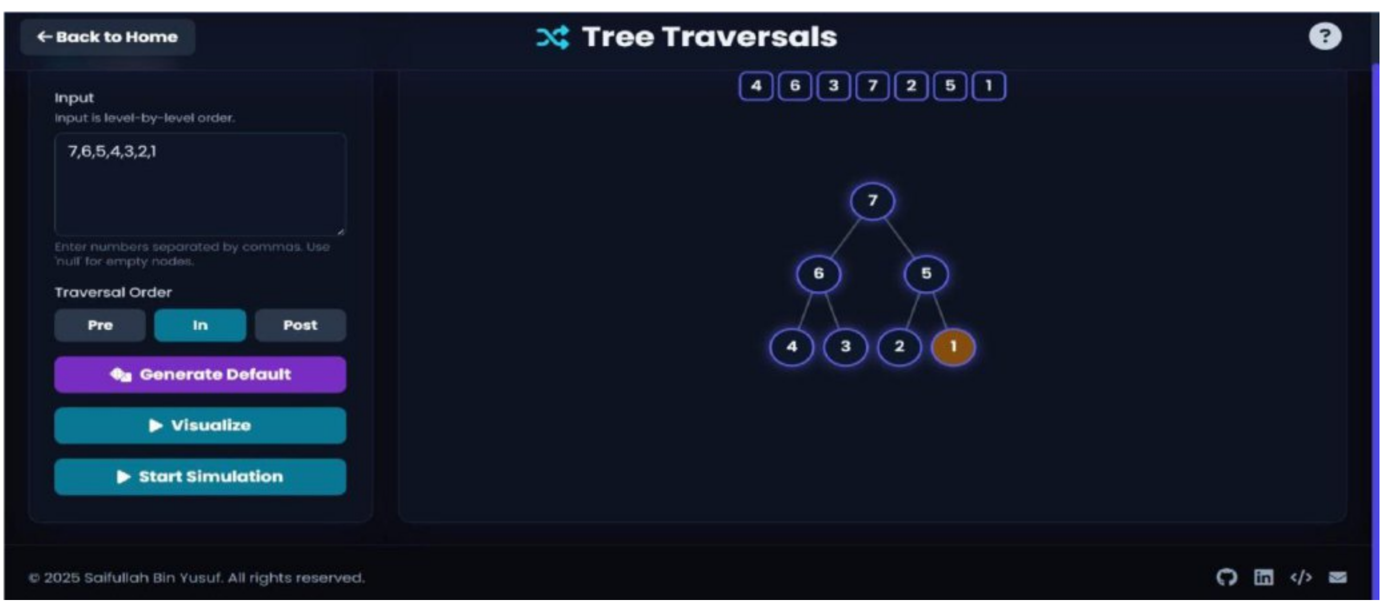
QUESTION-4



← Back to Home
Tree Traversals
4 3 6 2 1 5 7
Input
Input is level-by-level order.
7,6,5,4,3,2,1
Enter numbers separated by commas. Use 'null' for empty nodes.
Traversal Order
Pre
In
Post
Generate Default
Visualize
Start Simulation
7
6
5
4
3
2
1
© 2025 Saifullah Bin Yusuf. All rights reserved.

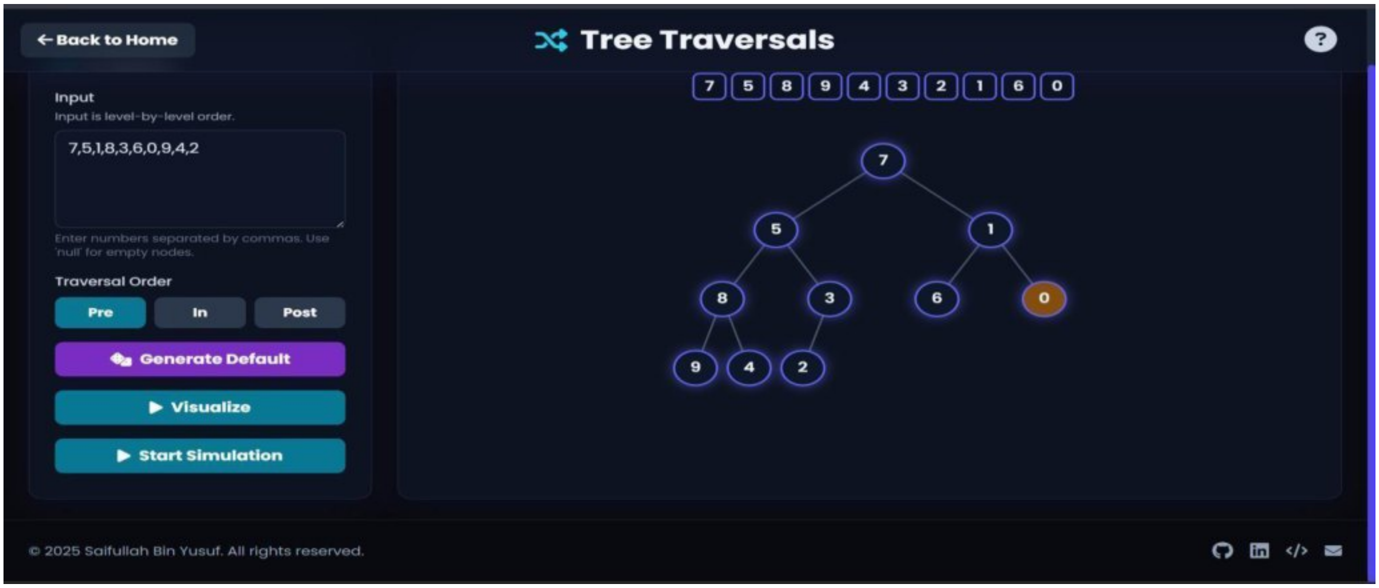


← Back to Home
Tree Traversals
7 6 4 3 5 2 1
Input
Input is level-by-level order.
7,6,5,4,3,2,1
Enter numbers separated by commas. Use 'null' for empty nodes.
Traversal Order
Pre
In
Post
Generate Default
Visualize
Start Simulation
7
6
5
4
3
2
1
© 2025 Saifullah Bin Yusuf. All rights reserved.

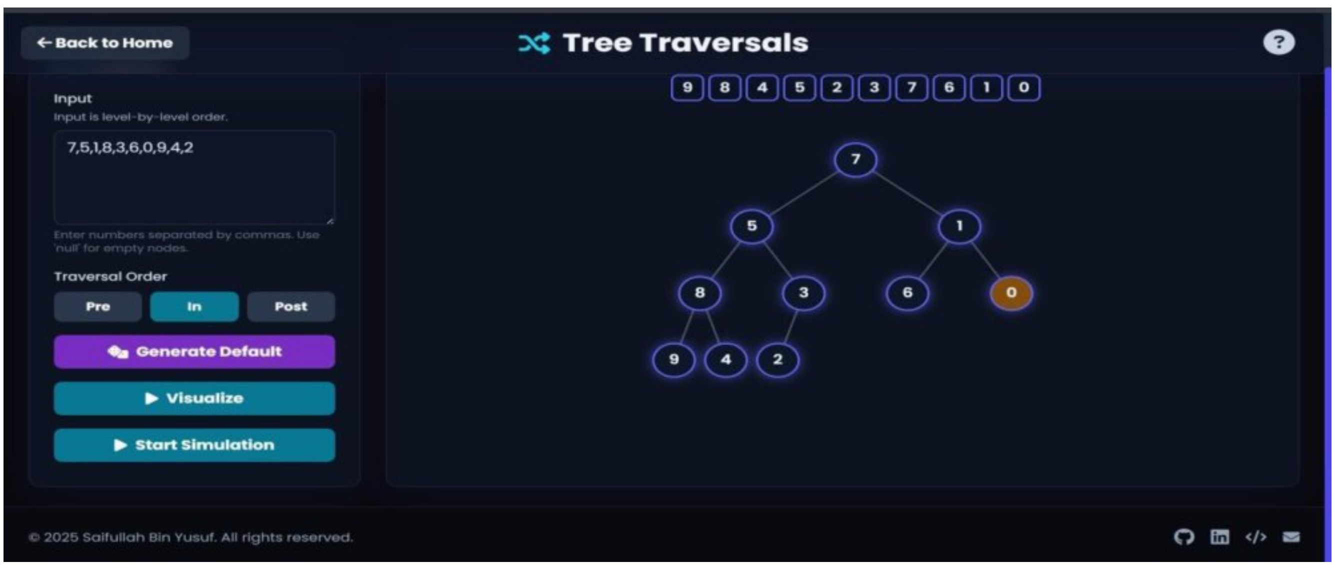


← Back to Home
Tree Traversals
4 6 3 7 2 5 1
Input
Input is level-by-level order.
7,6,5,4,3,2,1
Enter numbers separated by commas. Use 'null' for empty nodes.
Traversal Order
Pre
In
Post
Generate Default
Visualize
Start Simulation
7
6
5
4
3
2
1
© 2025 Saifullah Bin Yusuf. All rights reserved.

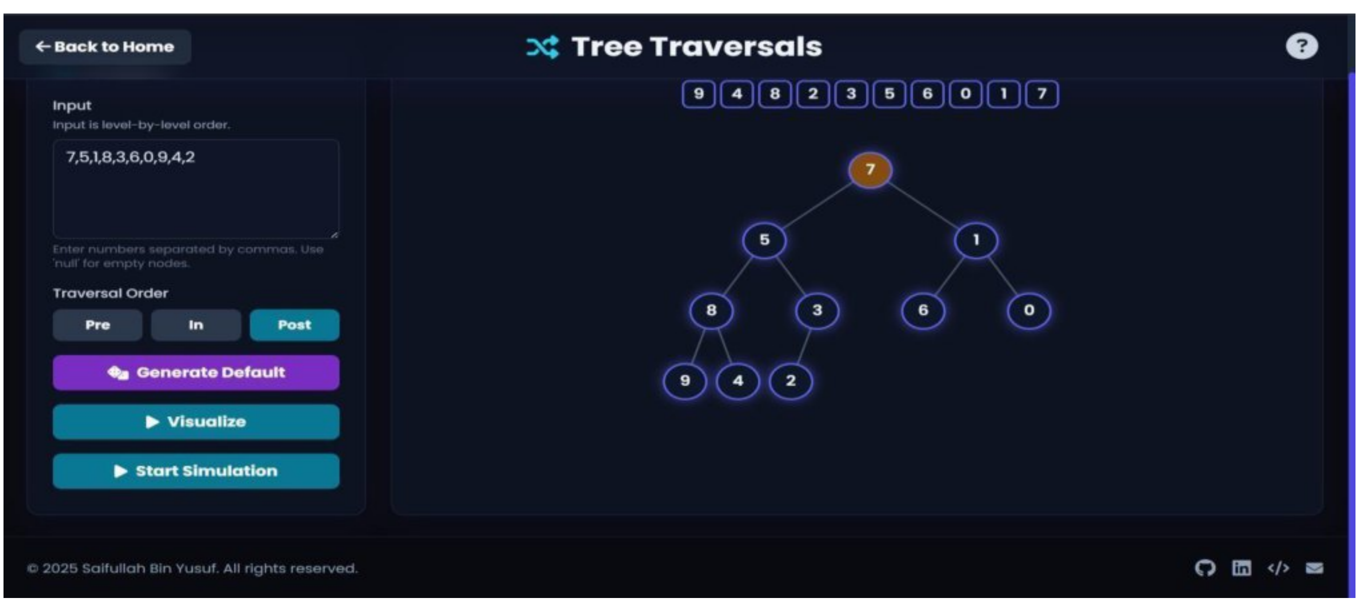
Question-5



← Back to Home
Tree Traversals
7 5 8 9 4 3 2 1 6 0
Input
Input is level-by-level order.
7,5,1,8,3,6,0,9,4,2
Enter numbers separated by commas. Use 'null' for empty nodes.
Traversal Order
Pre
In
Post
Generate Default
Visualize
Start Simulation
© 2025 Saifullah Bin Yusuf. All rights reserved.



← Back to Home
Tree Traversals
9 8 4 5 2 3 7 6 1 0
Input
Input is level-by-level order.
7,5,1,8,3,6,0,9,4,2
Enter numbers separated by commas. Use 'null' for empty nodes.
Traversal Order
Pre
In
Post
Generate Default
Visualize
Start Simulation
© 2025 Saifullah Bin Yusuf. All rights reserved.



← Back to Home
Tree Traversals
9 4 8 2 3 5 6 0 1 7
Input
Input is level-by-level order.
7,5,1,8,3,6,0,9,4,2
Enter numbers separated by commas. Use 'null' for empty nodes.
Traversal Order
Pre
In
Post
Generate Default
Visualize
Start Simulation
© 2025 Saifullah Bin Yusuf. All rights reserved.